A Geometric Control Approach for DC-Link Voltage Equalization in Split-Capacitor Inverters

Topic categories: **Modeling, Simulation, and Control; Multilevel Power Converters**

Abstract—Split-capacitor inverter topologies, such as the three-level neutral-point-clamped (NPC) distribution static compensator (DSTATCOM), suffer from major capacitor voltage imbalance when the neutral current contains a dc component, which degrades the current tracking performance of the compensator. A two-quadrant chopper is conventionally used to correct this imbalance, but its existing controllers switch continuously and respond slowly to a disturbance. In this paper, a simple geometric control approach inspired by boundary control theory is proposed to equalize the capacitor voltages. A second-order switching surface is derived from the chopper's own state trajectories, so that an imbalance is removed in a single switching action. The approach is applied to a three-level NPC DSTATCOM and validated through simulations, which show good transient behavior under dynamic loading with reduced switching.

I. INTRODUCTION

Voltage imbalance between split dc-link capacitors is a well-known problem in compensator topologies and may be classified as minor or major [1], [2]. Minor imbalance arises from practical factors such as unequal capacitances, leakage currents, asymmetrical current tracking and unequal device delays. The problem stays bounded when the neutral current entering the capacitor midpoint contains only ac components, which merely make the capacitor voltages oscillate, but a dc component makes them drift apart permanently, and this major imbalance degrades the current tracking of the compensator. Of the two main types of solution in literature, the first exploits the redundant switching states of the converter which lose authority at high modulation index and low power factor[3], and in a current-controlled compensator, are already dictated by the current controller, so such methods correct only minor imbalances [2]. The second adds a two-quadrant chopper that exchanges charge between the capacitors through an inductor connected to their midpoint [1], [4], controlled by hysteresis and by single-pulse, multi-pulse and PI duty-cycle schemes that solve its state equations repeatedly [1], and extended to diode-clamped multilevel inverters with bidirectional buck–boost choppers [5]. A carrier-based error control in discontinuous conduction removed these computations [2], but its carrier frequency must be derived in advance from the expected dc neutral current, and an improper choice leaves the inductor in mixed conduction with an unrestrained peak current. None of these schemes makes use of the natural trajectories of the chopper, which offer a way to minimize switching, improve the transient response and reduce the dependence on external parameters.

Boundary control, a nonlinear control technique widely used in power converters, selects a switching surface from the natural evolution of the converter states, and the behavior of the trajectories around this surface decides the transient response and stability [6], [7]. A first-order surface, as in sliding-mode control, offers good large-signal stability but a suboptimal transient response [8], [9], whereas a second-order surface derived from the natural trajectories brings the state to the target operating point in very few switching actions [9], usually at the cost of a peak current that is set by the system rather than by the designer.

This paper derives a second-order switching surface from the chopper's own state trajectories, inspired by boundary control. Since the surface follows the natural movement of the state, an imbalance is ideally removed in a single switching action, giving a fast transient response with minimal switching and a multi-step variant is introduced to limit the resulting high peak inductor current to an adjustable value. In a DSTATCOM (Fig. 4),

where the neutral current makes the capacitor voltages drift and oscillate within each cycle, two applications are proposed, the first countering both the drift and the oscillation, at the cost of a higher switching frequency and peak inductor current, and the second countering only the drift and leaving the bounded oscillation in place, with far fewer switching actions. The approaches are validated through simulations of a three-level NPC DSTATCOM, which agree with the theoretical predictions.

II. Proposed Control

A. Modelling and State Trajectories

Fig. 1 shows the two-quadrant chopper, where the switches S_1 and S_2 regulate the flow of charge between the capacitors C_1 and C_2 through an inductor L connected to their midpoint. Let the state of the chopper be $x = [e \;\; i_L]^T$, where $e = V_{C1} - V_{C2}$ is the voltage imbalance and i_L is the inductor current, taken positive into the midpoint. The chopper is first analyzed without a neutral current, which is later included in Section III, so the capacitors are balanced at $e = 0$ with no inductor current at rest, so the target operating point (OP) is $(e^*, i_L^*) = (0,0)$. For equal capacitances $C_1 = C_2 = C$ and a constant dc-link voltage $V_{in} = V_{C1} + V_{C2}$, (at least relatively constant over a few balancing cycles) the capacitor voltages are given by (1), and the KCL at the midpoint, $i_L = i_1 + i_2$, with the capacitor currents (2) gives the error dynamics (3).

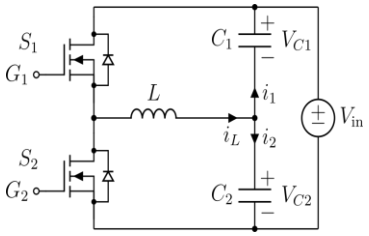


Fig. 1: Two-quadrant chopper

$$V_{C1} = \frac{V_{in}+e}{2}, \quad V_{C2} = \frac{V_{in}-e}{2} \quad (1) \qquad i_1 = -C\frac{dV_{C1}}{dt}, \quad i_2 = C\frac{dV_{C2}}{dt} \quad (2) \qquad C\frac{de}{dt} = -i_L \quad (3)$$

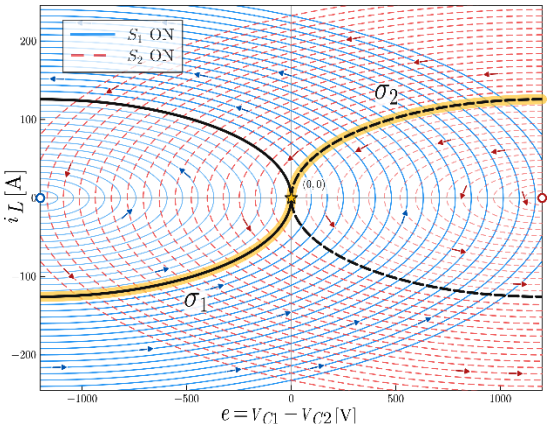


Fig. 2: State trajectories of the chopper

The inductor voltage (4) give the state equations (5), with the positive sign for S_1 ON and the negative sign for S_2 ON, and eliminating time gives the trajectory slope (6). Unlike the buck converter, whose second-order surface is derived by estimating the trajectory after a switching action [9], the slope equation (6) is variable-separable, so integrating it from the OP to (e, i_L) gives the exact trajectories through the target in closed form (7), with the same sign convention. The S_1 and S_2 trajectories form two families of concentric ellipses centred at $(-V_{in}, 0)$ and $(V_{in}, 0)$, as shown in Fig. 2. The best balancing strategy is to steer the state onto the trajectory through the OP on its own side, σ_2 for $e > 0$ and σ_1 for $e < 0$, and these two arcs, highlighted in Fig. 2, form the proposed switching surface (8).

$$L\frac{di_L}{dt} = V_{C1}\,;(S_1 \text{ on}), \;\; L\frac{di_L}{dt} = -V_{C2};(S_2 \text{ on}) \;(4) \qquad \frac{dx}{dt} = \begin{bmatrix} 0 & -\frac{1}{C} \\ \frac{1}{2L} & 0 \end{bmatrix} x \pm \begin{bmatrix} 0 \\ \frac{V_{in}}{2L} \end{bmatrix} \;(5) \qquad \frac{di_L}{de} = -\frac{C(e \pm V_{in})}{2Li_L} \;(6)$$

$$\sigma_{1,2}: \;\; i_L^2 + \frac{C}{2L}e^2 \pm \frac{CV_{in}}{L}e = 0 \;(7) \qquad \sigma(i_L, e) = \begin{cases} i_L^2 + \frac{C}{2L}e^2 - \frac{CV_{in}}{L}e, & e > 0, \\ i_L^2 + \frac{C}{2L}e^2 + \frac{CV_{in}}{L}e, & e < 0. \end{cases} \;(8)$$

B. Boundary Behavior and System Characteristics

The boundary analysis of a surface is a critical aspect of boundary control, since it decides the stability of the system. Reflective behavior around the OP is a necessary but not sufficient condition for stability [10], and the

proposed surface belongs to none of the classes in [7], [11], since the surface is itself made of S_1 and S_2 trajectories. On σ_2, the S_2 trajectory moves along the surface, while the S_1 trajectory differs from it only in di_L/dt, by the constant V_{in}/L, and so approaches the surface along i_L at this rate, and the same holds on σ_1 with S_1 and S_2 exchanged. No point of the surface is therefore rejective, every point is a transition point, and the state reaches the OP in a single switching action. The peak inductor current is another important system characteristic. Since i_L rises on the S_1 arc and falls along σ_2, it peaks at the switching point, and intersecting the S_1 trajectory through the initial state (e_0, i_{L0}) with σ_2 gives the error at this point (e_{sw}), and the peak current in closed form (9). For a start from rest with $|e_0| \ll V_{in}$, (9) reduces to $I_{pk} \approx \sqrt{CV_{in}|e_0|/(2L)}$, so the peak is set by the circuit and the initial imbalance, not by the controller. This single step control thus trades a high peak current for fast settling, and since many systems require a lower, adjustable current limit, a multi-step boundary control (Fig. 3) is proposed. It turns the switch off when i_L reaches a limit I_{peak} before the surface, lets the current freewheel to zero, and turns it on again from the new, smaller imbalance until the state meets the surface. The peak current therefore stays at I_{peak} and every turn-on occurs at zero current, at the cost of more switching actions to remove the same imbalance. In implementation, σ is fed to a relay with a small band $\pm b$, which sets the balancing tolerance, and a second relay on $|i_L|$ enforces I_{peak}.

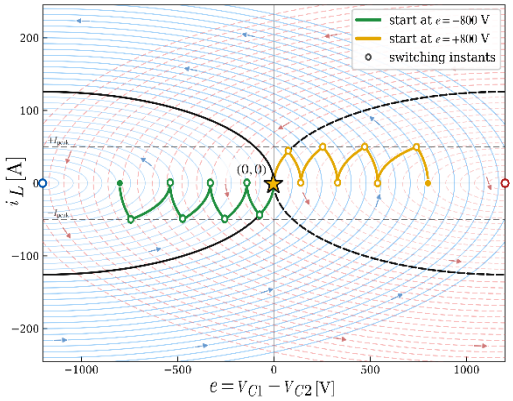


Fig. 3: Multi-step boundary control

$$e_{sw} = \frac{e_0}{2} + \frac{e_0^2}{4V_{in}} + \frac{Li_{L0}^2}{2CV_{in}} \;,\; I_{pk} = \sqrt{\frac{C}{2L} e_{sw}(2V_{in} - e_{sw})} \quad (9)$$

III. APPLICATION TO DSTATCOM

When the chopper is applied to a three-level NPC DSTATCOM (Fig. 4), unbalanced and nonlinear loads draw a neutral current i_o into the capacitor midpoint n', and this current contains a dc component, a supply-frequency component and its harmonics (10) [2]. Only the dc part $i_{o,\text{avg}}$, the mean over one supply period T, makes the capacitor voltages drift, while the ac part $i_{o,\text{ac}}$ only makes e oscillate within each cycle.

$$i_o = \mathrm{i}_{o,avg} + \sum_{h=1}^{\infty} I_h \sin(h\omega t + \phi_h) \quad (10) \qquad \sigma(i_L, e) = \begin{cases} \left(i_L + i_{o,\text{avg}}\right)^2 + \frac{C}{2L}e^2 - \frac{CV_{\text{in}}}{L}e, & e > 0, \\ \left(i_L + i_{o,\text{avg}}\right)^2 + \frac{C}{2L}e^2 + \frac{CV_{\text{in}}}{L}e, & e < 0. \end{cases} \quad (11)$$

Holding e at zero at every instant requires the inductor to carry the whole neutral current, which gives the target OP $(0, -i_o)$, whereas routing only its dc part through the inductor gives $\left(0, -i_{o,\text{avg}}\right)$, which is the aim of the chopper in [2]. The first target moves with $i_{o,\text{ac}}$, so the surface no longer matches the trajectories, the single switching action is lost, and the inductor current, which now follows the load, can neither be predicted by (9) nor limited by design. Choosing $\left(0, -i_{o,\text{avg}}\right)$ only shifts the state trajectories, so the analysis holds unchanged and the switching surface becomes (11), while $i_{o,\text{ac}}$ is countered naturally through error correction. With the multi-step limit, the steady state inductor current averages at most $I_{peak}/2$, so the drift is removed whenever $I_{peak} \geq 2\left|i_{o,\text{avg}}\right|$. This approach, where the inductor carries $i_{o,\text{avg}}$ while $i_{o,\text{ac}}$ is countered through error correction, is worthwhile

when the intra-cycle oscillation of e must be suppressed, for instance with a small dc-link capacitance, since its amplitude scales inversely with C. Where the oscillation stays within an allowable limit, it can be tolerated, and the ripple-compensated control below acts on the drift alone with far fewer switching actions.

$$e_r = \varepsilon - \frac{1}{T}\int_{t-T}^{t} \varepsilon \, d\tau, \quad \varepsilon(t) = -\frac{1}{C}\int_0^t i_{o,\mathrm{ac}} \, d\tau \quad \textbf{(12)} \qquad C\frac{d\hat{e}}{dt} = -\left(i_L + i_{o,\mathrm{avg}}\right) \quad \textbf{(13)}$$

A natural way to stop the chopper from reacting to the intra-cycle oscillation that $i_{o,\mathrm{ac}}$ causes in e is to low-pass filter e, but since the chopper corrects e on the same time scale as this oscillation, a filter slow enough to remove it would also delay the correction, and the surface would keep acting on an imbalance that has already been removed. The oscillation e_r is therefore predicted from the measured $i_{o,\mathrm{ac}}$ by (12) and subtracted from e, which removes it without delaying the chopper's own effect on e [12], [13. As (13) shows, the compensated error $\hat{e} = e - e_r$ sees only the dc part of the neutral current, so the chopper no longer acts on the intra-cycle oscillation, and since the prediction integrates the measured current, it holds for any periodic neutral current, harmonics included. This reduces the switching further, at the cost of a lag, as the one-cycle mean of i_o follows a change in its dc part only gradually.

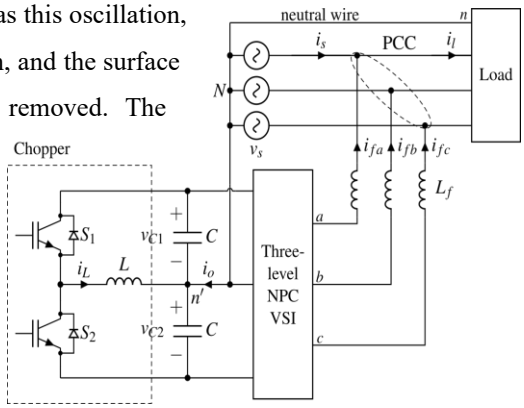


Fig. 4: Three-level NPC DSTATCOM schematic

IV.SIMULATION RESULTS

A standalone chopper with an induced disturbance was first simulated to verify the working of the control, which is detailed in the final paper. The approach was then applied to a three-level NPC DSTATCOM, with the system configuration taken from [2], a chopper inductance L = 100 mH, C = 2200 µF and V_{in}= 1200 V, which a PI controller regulates and the capacitors start balanced at V_{C1}= V_{C2} = 600 V. The direct application with single-step boundary control and the ripple-compensated application were verified first, and the system was then tested under dynamic loading with multi-step boundary control to observe its transient behavior. Fig. 5 shows the results of the direct application, where the capacitor voltages drift apart under the dc neutral current until the chopper is enabled at t = 0.5 s, and then equalize almost immediately, as seen in Fig. 5(a). In steady state, the oscillation common to V_{C1} and V_{C2} in Fig. 5(b) comes from V_{in}, which is regulated by the PI controller, while their difference stays within about ±0.4 V, a small residual that arises from the finite slope of the inductor current, as detailed in the final paper. Fig. 5(c) shows that the 25.2 V imbalance is removed in a single switching action, with the inductor current peaking at 18.1 A, in agreement with the 18.3 A predicted by (9), and the state settling within 5 ms. Finally, the mean of i_L in Fig. 5(d) matches that of i_o at 0.13 A, confirming that the inductor carries the dc neutral current, while $i_{o,\mathrm{ac}}$ is countered through error correction, at the cost of **1865 switching actions** per second.

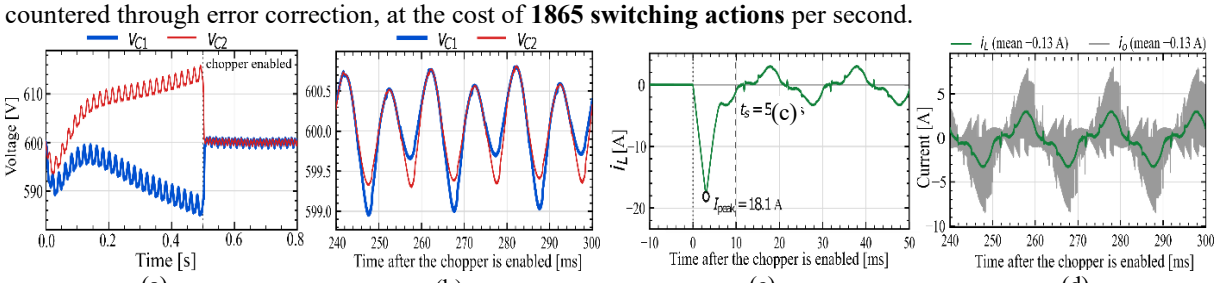


Figure 5: Voltage and current waveforms under direct application of single step boundary

Fig. 6 shows the same run with the ripple-compensated application, where the chopper removes the drift in Fig. 6(a) as before but, acting on $\hat{e}$, leaves the bounded intra-cycle oscillation in place. In Fig. 6(b), e reaches about ±3.3 V, while the capacitor means differ by only 0.2 V, an offset set by the relay band. The peak in Fig. 6(c) rises to 19.1 A, since the surface now acts on $\hat{e}$, which is 27.9 V at enabling, in agreement with the 19.3 A predicted by (9). In steady state, the dc neutral current drives $\hat{e}$ in one direction, so the chopper fires a one-sided pulse each time $\hat{e}$ reaches the band edge, and Fig. 6(d) shows the same mean current as before with only **140 switching actions** per second against 1865, about 13 times fewer.

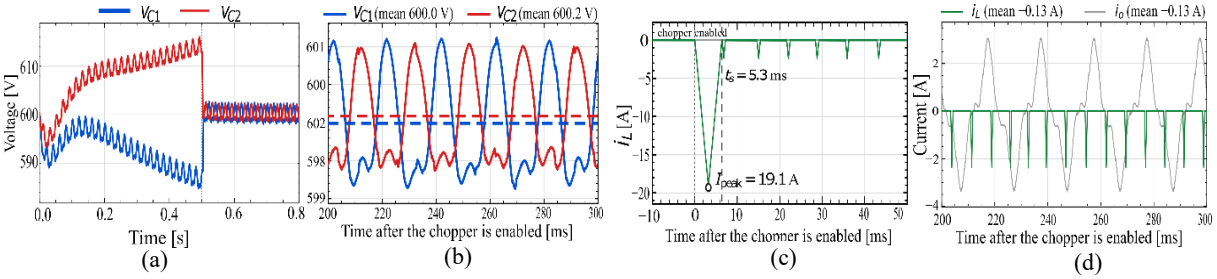


Figure 6: Voltage and current waveforms under ripple compensated application of single step boundary

The most important aspect to verify is the transient behavior of the converter, since a fast transient response is the main advantage of boundary control. The ripple-compensated application is therefore simulated with multi-step control and two step changes in the load, made by connecting and then reversing two half-wave rectifiers at t = 0.5 s and t = 0.7 s. The multi-step control is validated in Fig. 7(b), where, with I_{peak} set at 6 A, the chopper takes several current pulses to reach the switching surface and settles within 10.4 ms, longer than a single switching action, as expected for the lower peak current. The load steps then reverse the dc neutral current twice from -0.13A to 1.14 A, and then to -1.39 A, a severe test for any balancing scheme. The chopper follows each change within a few supply cycles in Fig. 7(c), with only a brief overshoot while the one-cycle average of the neutral current catches up, and the capacitor voltages in Fig.7(a) remain balanced throughout. These results validate the proposed control under dynamic loading, and its finer details are discussed in the final paper.

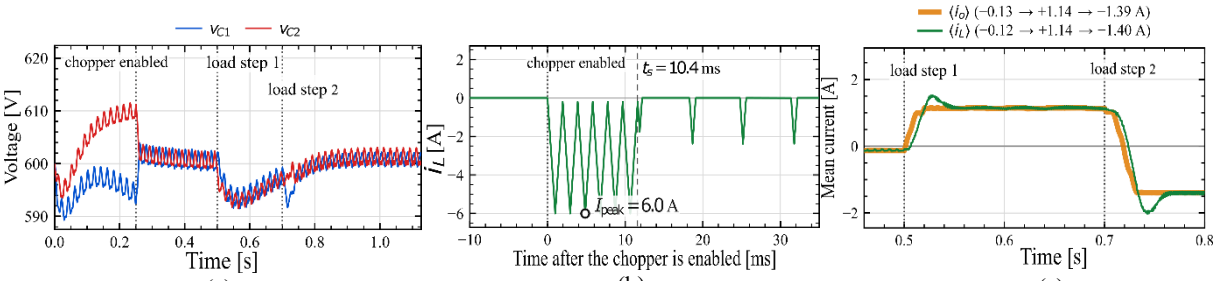


Figure 7: Voltage and current waveforms in ripple compensated multi-step boundary control during dynamic loading

CONCLUSION

This work has shown that a two-quadrant chopper can equalize the dc-link capacitor voltages through its own natural dynamics, when its switching follows a second-order surface built from its state trajectories. The resulting fast transient response with minimal switching comes at the cost of a higher peak current, though predictable, peak current, which a multi-step variant limits to an adjustable value. When applied to a three-level NPC DSTATCOM, the ripple compensated approach further reduces switching by tolerating the bounded intra-cycle oscillations. The final paper will include a detailed analysis of the control, a comparative study and experimental validation.

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